



2019 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen, black is preferred
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks: 100

Section I – 10 marks (pages 2 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8 – 17)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer page in the writing booklet for Questions 1 – 10.

1 What is the value of i^{2019} ?

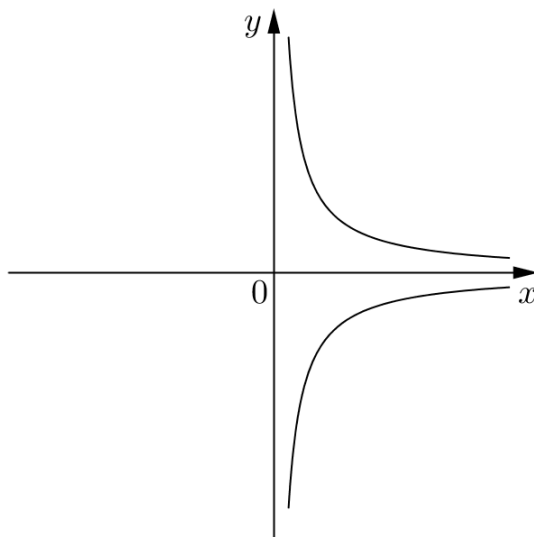
A. i

B. $-i$

C. 1

D. -1

2 Which of the equations below best represents the following graph?



A. $|x|y = 1$

B. $x|y| = 1$

C. $|xy| = 1$

D. $x^2y^2 = 1$

3 What are the equations of the asymptotes of the hyperbola $4x^2 - 25y^2 = 100$?

A. $y = \pm \frac{2}{5}x$

B. $y = \pm \frac{4}{25}x$

C. $y = \pm \frac{5}{2}x$

D. $y = \pm \frac{25}{4}x$

4 Multiplying a non-zero complex number by $\frac{1+i}{1-i}$ results in a rotation about the origin on an Argand diagram. What is the rotation?

A. Clockwise by $\frac{\pi}{2}$ radians

B. Clockwise by $\frac{\pi}{4}$ radians

C. Anticlockwise by $\frac{\pi}{2}$ radians

D. Anticlockwise by $\frac{\pi}{4}$ radians

5 The polynomial equation $x^3 - 3x^2 - x + 2 = 0$ has roots α, β and γ . Which of the following polynomial equations has roots $2\alpha + \beta + \gamma$, $\alpha + 2\beta + \gamma$ and $\alpha + \beta + 2\gamma$?

A. $x^3 - 6x^2 - 2x + 4 = 0$

B. $x^3 + 6x^2 + 8x - 1 = 0$

C. $x^3 + 12x^2 - 44x + 49 = 0$

D. $x^3 - 12x^2 + 44x - 49 = 0$

- 6 The motion of a particle undergoing simple harmonic motion is oscillatory, of which the amplitude of the motion remains constant.

However, in many real-world applications, oscillatory motions are often dampened out of necessity. The motion of a particle undergoing dampened harmonic motion is still oscillatory, but the amplitude of its motion reduces over time.

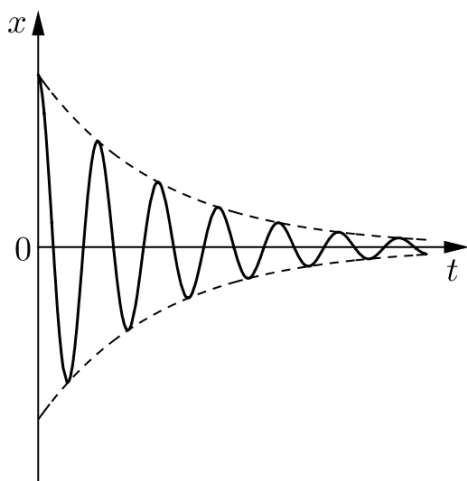
One such motion can be modelled by the equation:

$$x = e^{-\gamma t} \cos(\omega t)$$

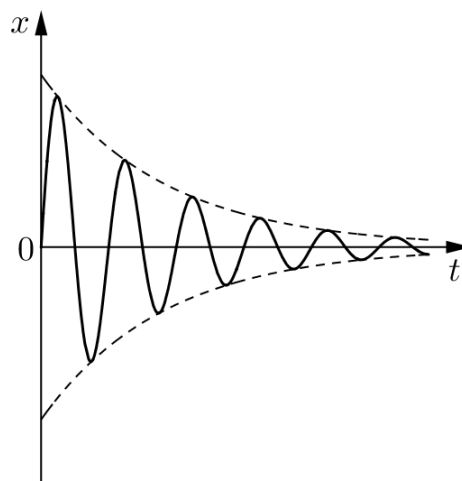
where γ and ω are positive constants, and x is the displacement of the particle from the centre of its oscillatory motion over time t .

Which of the following graphs best describes this model?

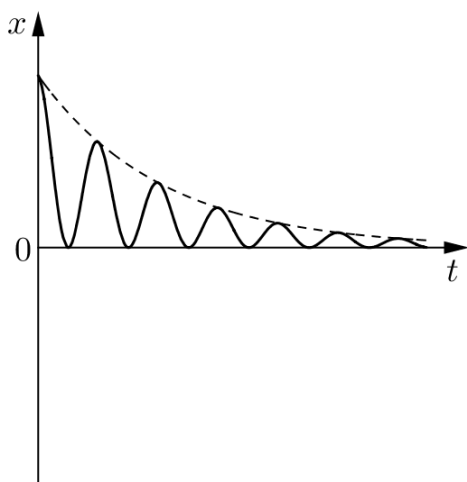
A.



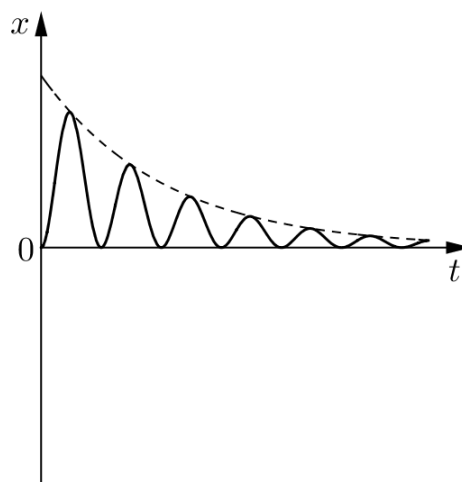
B.



C.

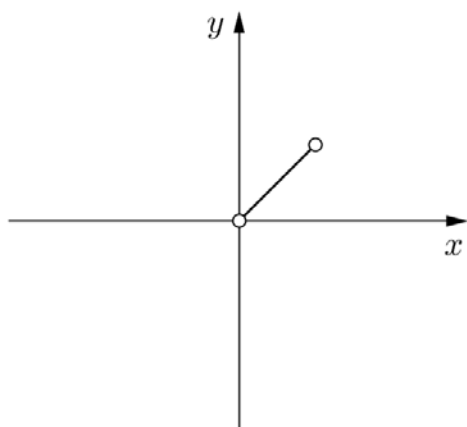


D.

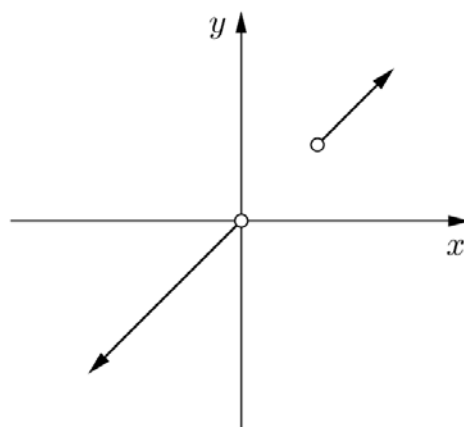


7 Which diagram best represents the solutions to the equation $\arg(z) = \arg(z + 2 + 2i)$?

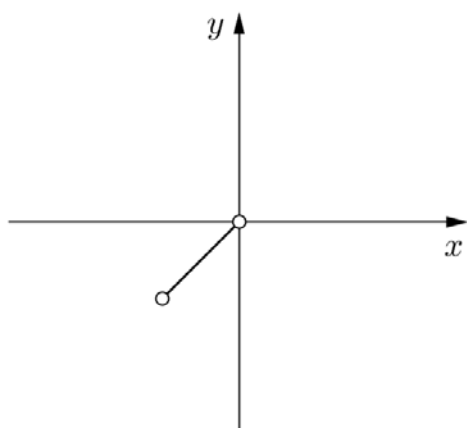
A.



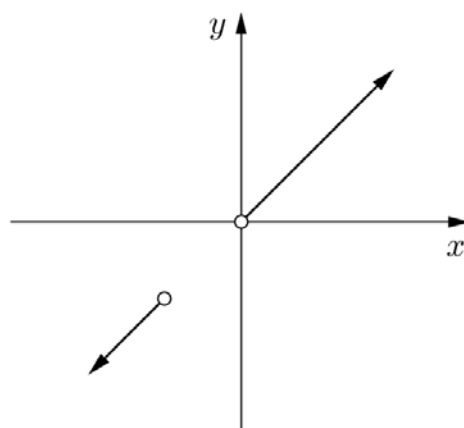
B.



C.



D.



8 It is given that $3+i$ is a root of $P(z) = z^3 + az^2 + bz + 10$ where a and b are real numbers. Which of the following expressions for $P(z)$ is correct?

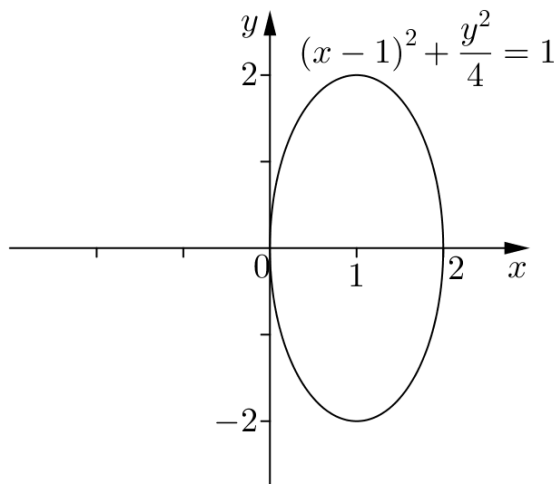
A. $P(z) = (z-1)(z^2 + 6z - 10)$

B. $P(z) = (z-1)(z^2 - 6z - 10)$

C. $P(z) = (z+1)(z^2 + 6z + 10)$

D. $P(z) = (z+1)(z^2 - 6z + 10)$

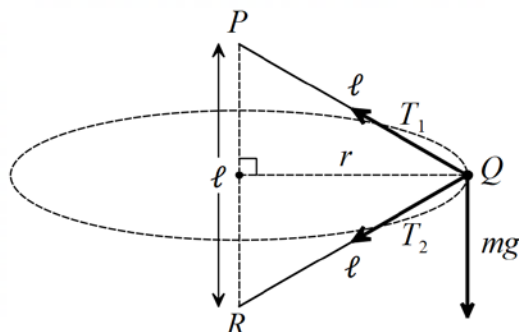
- 9 The region enclosed by the ellipse $(x-1)^2 + \frac{y^2}{4} = 1$ is rotated about the y -axis to form a solid.



Which integral represents the volume of the solid formed?

- A. $\int_{-2}^2 \pi \sqrt{4-y^2} dy$
- B. $\int_{-2}^2 \pi \sqrt{1-y^2} dy$
- C. $\int_{-2}^2 2\pi \sqrt{4-y^2} dy$
- D. $\int_{-2}^2 2\pi \sqrt{1-y^2} dy$

- 10 Two light inextensible strings PQ and QR each of length ℓ are attached to a particle of mass m at Q . The other ends P and R are fixed to two points in a vertical line such that P is at a distance ℓ above R . The particle moves in a horizontal circle with constant angular velocity ω such that both strings remain taut.



Taking $g \text{ m/s}^2$ as the acceleration due to gravity, what are the tensions in the strings?

- A. $T_1 = \frac{m}{2}(\ell\omega^2 + 2g)$ and $T_2 = \frac{m}{2}(\ell\omega^2 - 2g)$
- B. $T_1 = \frac{m}{2}(\ell\omega^2 - 2g)$ and $T_2 = \frac{m}{2}(\ell\omega^2 + 2g)$
- C. $T_1 = m(\ell\omega^2 + 2g)$ and $T_2 = m(\ell\omega^2 - 2g)$
- D. $T_1 = m(\ell\omega^2 - 2g)$ and $T_2 = m(\ell\omega^2 + 2g)$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate section of the writing booklet. Extra writing paper is available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 section of the writing booklet.

- (a) Given $z = 3 + i$ and $w = 1 - 2i$, express $\frac{5z}{w}$ in $x + iy$ form, where x and y are real numbers. 2
- (b) (i) Write $\sqrt{3} + i$ in modulus-argument form. 2
- (ii) Find the smallest positive integer n such that $(\sqrt{3} + i)^n$ is a real number. 1
- (c) (i) Find real numbers a , b and c such that $\frac{4}{x^2(2-x)} \equiv \frac{ax+b}{x^2} + \frac{c}{2-x}$. 2
- (ii) Hence, or otherwise, find $\int \frac{4}{x^2(2-x)} dx$. 3
- (d) Find $\int \frac{dx}{\sqrt{1-4x-x^2}}$. 2
- (e) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$ using the substitution $t = \tan \frac{x}{2}$. 3

End of Question 11

Question 12 (15 marks) Use the Question 12 section of the writing booklet.

(a) The equation $x^3 + x^2 + 2 = 0$ has roots α, β and γ .

(i) Show that $\alpha^2 + \beta^2 + \gamma^2 = 1$. 2

(ii) Evaluate $\alpha^3 + \beta^3 + \gamma^3$. 2

(iii) Evaluate $\alpha^4 + \beta^4 + \gamma^4$. 2

(b) (i) Sketch the region on the Argand diagram where the following inequalities hold simultaneously. 3

$$|z - 1| \leq \sqrt{2} \quad \text{and} \quad 0 \leq \arg(z + i) \leq \frac{\pi}{4}$$

(ii) Let w be the complex number of minimum modulus satisfying the inequalities in part (i). Express w in the form $x + iy$, where x and y are real number. 1

(c) How many thirteen-letter arrangements can be made using the letters of the word BAULKHAMHILLS if:

(i) there are no restrictions? 1

(ii) all seven of the repeated letters are grouped together in no particular order? 2

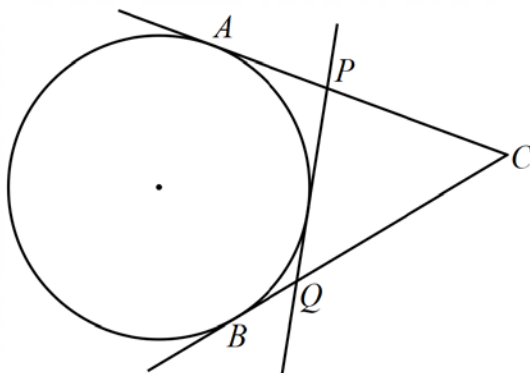
(iii) the vowels appear in alphabetical order but are separated by at least one consonant? 2

End of Question 12

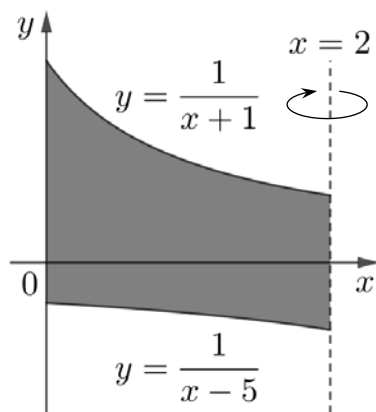
Question 13 (15 marks) Use the Question 13 section of the writing booklet.

- (a) A and B are two points on a circle. Tangents at A and B meet at C . A third tangent on the minor arc AB cuts CA and CB at P and Q respectively, as shown in the diagram below. 2

Copy the diagram into your writing booklet and show that the perimeter of $\triangle CPQ$ is independent of PQ .



- (b) The region between the curves $y = \frac{1}{x+1}$, $y = \frac{1}{x-5}$, the y -axis and the line $x = 2$ is rotated about the line $x = 2$ to form a solid.

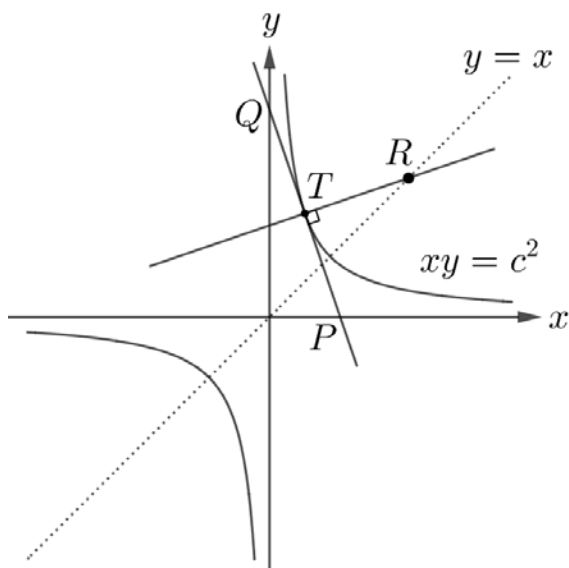


- (i) Using the method of cylindrical shells, show that the volume of a typical shell of thickness Δx is given by $\Delta V = 12\pi \left(\frac{x-2}{x^2-4x-5} \right) \Delta x$. 2
- (ii) Hence, or otherwise, find the exact volume of the resulting solid. 3

Question 13 continues on the next page

Question 13 (continued)

- (c) The point $T\left(ct, \frac{c}{t}\right)$ lies on the hyperbola $xy = c^2$. The tangent at T meets the x -axis at P and the y -axis at Q . The normal at T meets the line $y = x$ at R .



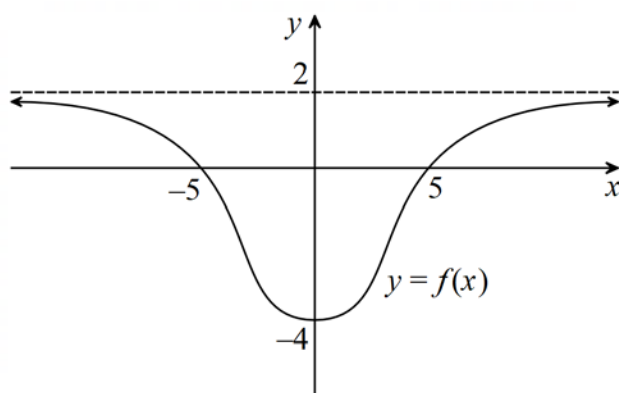
You may assume that the tangent at T has equation $x + t^2y = 2ct$ (Do NOT prove this.)

- | | |
|--|---|
| (i) Find the coordinates of P and Q . | 2 |
| (ii) Find the equation of the normal at T . | 2 |
| (iii) Show that the x -coordinate of R is $\frac{c}{t}(t^2 + 1)$. | 2 |
| (iv) Prove that $\triangle PQR$ is isosceles. | 2 |

End of Question 13

Question 14 (15 marks) Use the Question 14 section of the writing booklet.

(a) The diagram shows the graph of $y = f(x)$.



Draw sketches of the graphs of the following on separate number planes:

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = [f(x)]^2$ 2

(iii) $y^2 = -f(x)$ 2

(b) A particle of mass m kilograms is projected vertically upwards through a medium under the influence of gravity with an initial velocity of U m/s. In this medium, the particle experiences a resistive force of mkv newtons, where k is a positive constant and v is the particle's velocity in m/s. The acceleration due to gravity for this motion is g m/s².

It is known that the particle reaches its greatest height H metres in T seconds.

(i) Show that $T = \frac{1}{k} \ln \left(\frac{g + kU}{g} \right)$. 2

(ii) Show that $\frac{dx}{dv} = -\frac{v}{g + kv}$, where x is the particle's displacement from its point of projection in metres. 1

(iii) Hence, or otherwise, show that $U = kH + gT$. 2

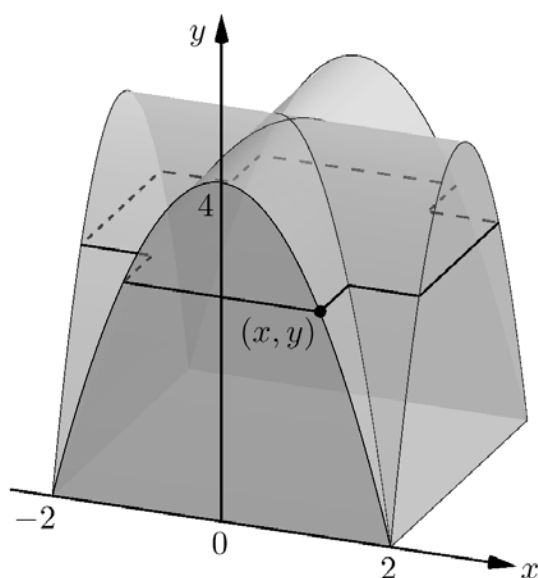
Question 14 continues on the next page

Question 14 (continued)

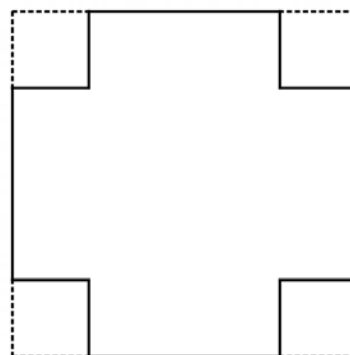
- (c) A sandstone cap on the corner of a fence is shown below. The solid is formed by intersecting two parabolic cylinders (a prism with uniform parabolic cross-sections).

On the front face, the equation of the parabola is $y = 4 - x^2$.

Slices parallel to the base of the solid are in the shape of a square with four smaller squares removed, one from each corner. The shape of a typical horizontal slice at height y is also shown below.



Sandstone cap



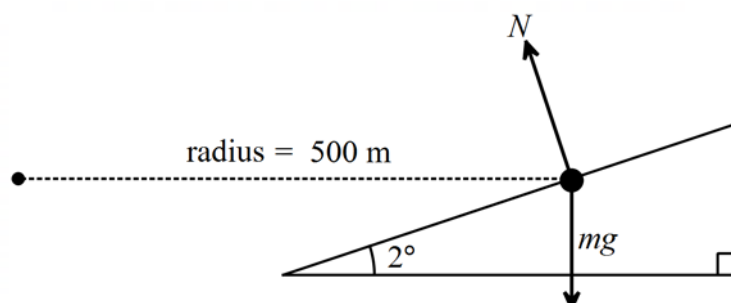
Shape of a typical slice

- (i) Show that the area of a typical slice is $16 - 4 \times (2 - \sqrt{4 - y})^2$ square units. 1
- (ii) Hence find the volume of the sandstone cap. 3

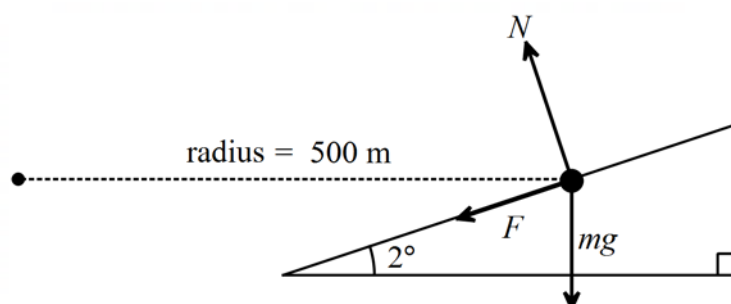
End of Question 14

Question 15 (15 marks) Use the Question 15 section of the writing booklet.

- (a) A 5000 kg truck is travelling around a circular portion of road of radius 500 metres. This portion of the road is banked at an angle of 2° to the horizontal. A cross-section of the road is shown in the diagram below. (Take g to be 10 m/s^2 .)



- (i) By resolving forces, determine the speed at which the truck must negotiate this circular portion of road such that the truck experiences no sideways friction. 2
- (ii) If the truck travels around this circular portion of road at a speed of 72 km/h, how much sideways friction (in newtons) is exerted on the tyres of the truck? 3



- (b) (i) Show that $\tan\left(A + \frac{\pi}{2}\right) = -\cot A$. 1
- (ii) Use mathematical induction to prove that $\tan\left[(2n+1)\frac{\pi}{4}\right] = (-1)^n$ for all integers $n \geq 1$. 3

Question 15 continues on the next page

Question 15 (continued)

(c) Let $I_n = \int_0^1 \frac{x^n}{\sqrt{x+1}} dx$, where n is an integer and $n \geq 0$.

(i) Show that $I_0 = 2\sqrt{2} - 2$. **1**

(ii) Show that $I_{n-1} + I_n = \int_0^1 x^{n-1} \sqrt{x+1} dx$. **1**

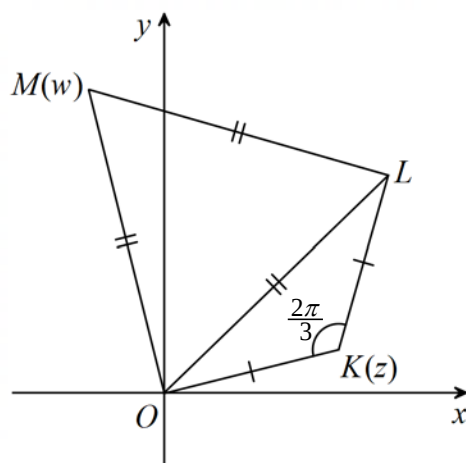
(iii) Use integration by parts to show that $I_n = \frac{2\sqrt{2} - 2nI_{n-1}}{2n+1}$ where n is an integer and $n \geq 1$. **2**

(iv) Hence, or otherwise, evaluate $\int_0^1 \frac{x^2}{\sqrt{x+1}} dx$. **2**

End of Question 15

Question 16 (15 marks) Use the Question 16 section of the writing booklet.

- (a) On the Argand diagram below, the complex numbers z and w are represented by the points K and M respectively. It is given that $\triangle OKL$ is isosceles with $\angle OKL = \frac{2\pi}{3}$ as shown, and $\triangle OLM$ is equilateral. All angles are measured in radians.

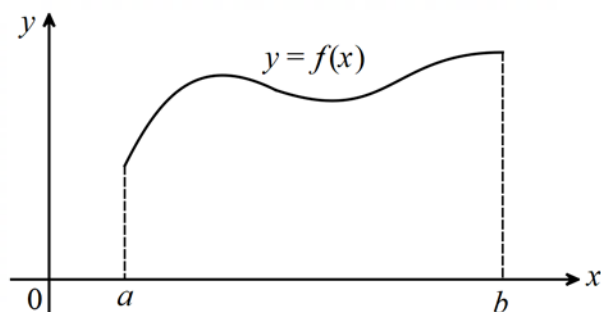


- | | |
|--|----------|
| (i) Show that $\angle MOK = \frac{\pi}{2}$. | 1 |
| (ii) Show that $ \overrightarrow{OL} = \sqrt{3} \times z $. | 2 |
| (iii) Hence show that $3z^2 + w^2 = 0$. | 2 |

Question 16 continues on the next page

Question 16 (continued)

- (b) The curve below represents $y = f(x)$, which is continuous and differentiable in the domain $a \leq x \leq b$.



The length of this curve, L , from $x = a$ to $x = b$ is given by $L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$.

Now consider the curve $y = \frac{1}{2}(e^x + e^{-x})$.

- (i) Show that, for this curve, $1 + \left(\frac{dy}{dx}\right)^2 = \frac{(e^x + e^{-x})^2}{4}$. 2
- (ii) Hence calculate the exact length of this curve from $x = 0$ to $x = 1$. 2
- (c) A polynomial of degree n is given by $P(x) = x^n + px - q$. It is given that the polynomial has a double root at $x = \alpha$.
- (i) Show that $\alpha^{n-1} = -\frac{p}{n}$. 2
- (ii) Show that $\left(\frac{p}{n}\right)^n + \left(\frac{q}{n-1}\right)^{n-1} = 0$. 3
- (iii) Hence, or otherwise, deduce that the double root is $\frac{qn}{p(n-1)}$. 1

End of Paper



YEAR 12 TRIAL EXAMINATION 2019
MATHEMATICS EXTENSION 2
MARKING GUIDELINES

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	B
3	A
4	C
5	D

Question	Answer
6	A
7	D
8	D
9	C
10	A

Questions 1 – 10

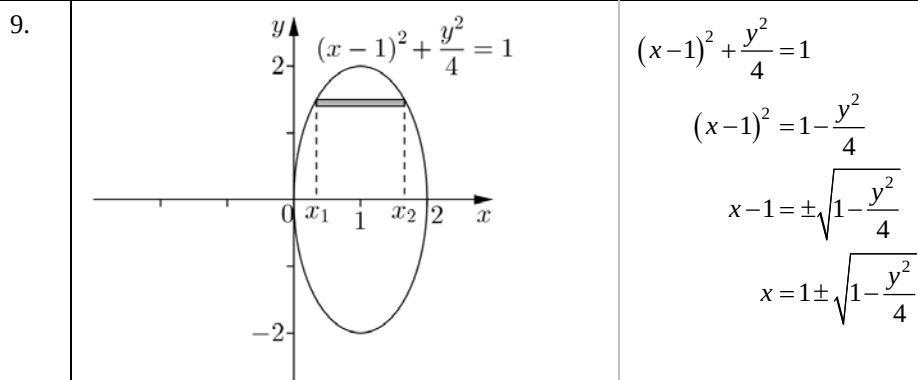
Sample solution			
1.	$i^{2019} = \left(i^4\right)^{504} \times i^3$ $= 1^{504} \times (-i)$ $= -i$		
2.	The given graph is the portion of graph $xy = 1$ above the x -axis as well as its reflection in the x -axis, so replace y in the equation with $ y $. The equation of the given graph is therefore $x y = 1$		
3.	$4x^2 - 25y^2 = 100$ $\frac{x^2}{25} - \frac{y^2}{4} = 1$	$a^2 = 25 \Rightarrow a = 5$ $b^2 = 4 \Rightarrow b = 2$	Asmyptotes are $y = \pm \frac{b}{a}x$, i.e. $y = \pm \frac{2}{5}x$
4.	$\frac{1+i}{1-i} = \frac{\sqrt{2}\left(\text{cis}\frac{\pi}{4}\right)}{\sqrt{2}\text{cis}\left(-\frac{\pi}{4}\right)}$ $= \text{cis}\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right]$ $= \text{cis}\frac{\pi}{2}$	Multiplying by $\frac{1+i}{1-i}$ is therefore equivalent to multiplying by $\text{cis}\frac{\pi}{2}$, which causes an anticlockwise rotation by $\frac{\pi}{2}$ radians.	
5.	For the new polynomial, $\sum \alpha = (2\alpha + \beta + \gamma) + (\alpha + 2\beta + \gamma) + (\alpha + \beta + 2\gamma)$ $= 4\alpha + 4\beta + 4\gamma$ $= 4(\alpha + \beta + \gamma)$ $= 4 \times 3$ $= 12$ Only option D satisfies this property.		
6.	$-1 \leq \cos(\omega t) \leq 1$ $-e^{-\gamma t} \leq e^{-\gamma t} \cos(\omega t) \leq e^{-\gamma t}$ When $t = 0$, $e^0 \cos 0 = 1$ Option A satisfies both of these properties.		

Sample solution

7. $\arg(z) = \arg(z + 2 + 2i)$
 $\arg(z) = \arg[z - (-2 - 2i)]$
 Critical points are therefore $z = 0$ and $z = -2 - 2i$.
 Option D is the best representation of the solutions to the equation $\arg(z) = \arg(z + 2 + 2i)$

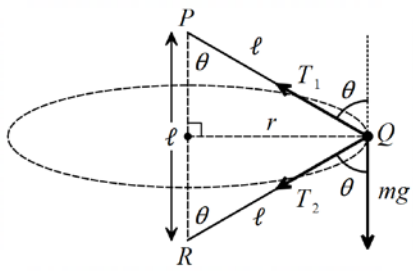
8. Since a and b are real numbers, and $3 + i$ is a root of $P(z)$, therefore $3 - i$ must also be a root of $P(z)$.
 Therefore, $[z - (3 + i)][z - (3 - i)]$ is a factor of $P(z)$.

$$\begin{aligned} [z - (3 + i)][z - (3 - i)] &= (z - 3 - i)(z - 3 + i) \\ &= (z - 3)^2 - i^2 \\ &= z^2 - 6z + 9 - (-1) \\ &= z^2 - 6z + 10 \end{aligned}$$



$$\begin{aligned} \Delta V &= \pi(x_2^2 - x_1^2) \Delta y \\ &= \pi \left[\left(1 + \sqrt{1 - \frac{y^2}{4}} \right)^2 - \left(1 - \sqrt{1 - \frac{y^2}{4}} \right)^2 \right] \Delta y \\ &= \pi \left(2 \times 2 \sqrt{1 - \frac{y^2}{4}} \right) \Delta y \\ &= 4\pi \sqrt{1 - \frac{y^2}{4}} \Delta y \end{aligned}$$

$$\begin{aligned} V &= \lim_{\Delta y \rightarrow 0} \sum_{y=-2}^2 4\pi \sqrt{1 - \frac{y^2}{4}} \Delta y \\ &= \int_{-2}^2 4\pi \sqrt{1 - \frac{y^2}{4}} dy \\ &= \int_{-2}^2 2\pi \sqrt{4 - y^2} dy \end{aligned}$$

10. 
- ΔPQR is equilateral, $\therefore \cos \theta = \cos 60^\circ = \frac{1}{2}$
 $\sin \theta = \frac{r}{\ell}$
- Resolving vertically:
 $T_1 \cos \theta - T_2 \cos \theta - mg = 0$

$$T_1 - T_2 = \frac{mg}{\cos \theta}$$

$$T_1 - T_2 = 2mg$$
- Resolving horizontally:
 $T_1 \sin \theta + T_2 \sin \theta = mr\omega^2$

$$T_1 + T_2 = \frac{mr\omega^2}{\sin \theta}$$

$$= m\ell\omega^2$$

Solving simultaneously,

$$\begin{aligned} T_1 &= \frac{m\ell\omega^2 + 2mg}{2} & T_2 &= \frac{m\ell\omega^2 - 2mg}{2} \\ &= \frac{m}{2}(\ell\omega^2 + 2g) & &= \frac{m}{2}(\ell\omega^2 - 2g) \end{aligned}$$

Section II

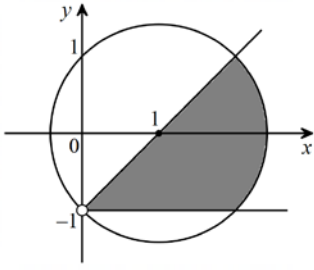
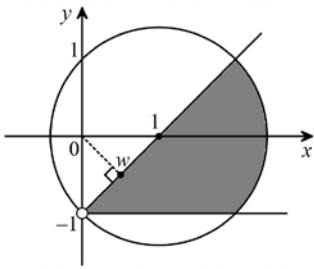
Question 11

Sample solution	Suggested marking criteria
(a) $\frac{5z}{w} = \frac{5(3+i)}{1-2i}$ $= \frac{15+5i}{1-2i}$ $= \frac{(15+5i)(1+2i)}{(1-2i)(1+2i)}$ $= \frac{15+30i+5i+10i^2}{1^2-(2i)^2}$ $= \frac{5+35i}{5}$ $= 1+7i$	• 2 – correct solution • 1 – realises the denominator
(b) (i) $\sqrt{3}+i = 2 \operatorname{cis} \frac{\pi}{6}$	• 2 – correct solution • 1 – correct modulus – correct argument (not necessarily principal argument)
(ii) $(\sqrt{3}+i)^n = \left(2 \operatorname{cis} \frac{\pi}{6}\right)^n$ $= 2^n \operatorname{cis} \frac{n\pi}{6}$ This is a real number if the argument is an integer multiple of π, the smallest positive integer n such that this happens is $n = 6$.	• 1 – correct solution
(c) (i) $\frac{4}{x^2(2-x)} \equiv \frac{ax+b}{x^2} + \frac{c}{2-x}$ $4 \equiv (ax+b)(2-x) + cx^2$ $4 \equiv 2ax - ax^2 + 2b - bx + cx^2$ $4 \equiv (c-a)x^2 + (2a-b)x + 2b$ Comparing coefficients: $\begin{array}{lll} 2b = 4 & 2a - b = 0 & c - a = 0 \\ b = 2 & 2a - 2 = 0 & c - 1 = 0 \\ & 2a = 2 & c = 1 \\ & a = 1 & \end{array}$	• 2 – correct solution • 1 – correctly finds two of the three coefficients
(ii) $\int \frac{4}{x^2(2-x)} dx = \int \frac{x+2}{x^2} + \frac{1}{2-x} dx$ $= \int \frac{1}{x} + 2x^{-2} - \frac{1}{2-x} dx$ $= \ln x - \frac{2}{x} - \ln(2-x) + c$ $= \ln \frac{x}{2-x} - \frac{2}{x} + c$	• 3 – correct solution • 2 – correctly integrates two of the three integrals listed below, or equivalent merit • 1 – correctly integrates one of the three integrals ($\frac{1}{x}$, $2x^{-2}$ or $\frac{1}{2-x}$), or equivalent merit

Question 11 (continued)

Sample solution	Suggested marking criteria
(d) $\int \frac{dx}{\sqrt{1-4x-x^2}} = \int \frac{dx}{\sqrt{-(x^2+4x-1)}}$ $= \int \frac{dx}{\sqrt{-(x+2)^2-5}}$ $= \int \frac{dx}{\sqrt{5-(x+2)^2}}$ $= \sin^{-1}\left(\frac{x+2}{\sqrt{5}}\right) + c$	• 2 – correct solution • 1 – attempts to complete the square and reduces the integral to a “standard” integral
(e) $t = \tan \frac{x}{2}$ $\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$ $= \frac{1}{2} \left(1 + \tan^2 \frac{x}{2}\right)$ $= \frac{1+t^2}{2}$ $\frac{dx}{dt} = \frac{2}{1+t^2}$	When $x = 0, t = 0$. When $x = \frac{\pi}{2}, t = 1$. $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x} = \int_0^1 \frac{\frac{2dt}{1+t^2}}{2 + \frac{1-t^2}{1+t^2}}$ $= \int_0^1 \frac{2dt}{2(1+t^2) + (1-t^2)}$ $= \int_0^1 \frac{2dt}{3+t^2}$ $= 2 \times \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$ $= 2 \times \left(\frac{1}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{3}} \tan^{-1} 0 \right)$ $= \frac{2}{\sqrt{3}} \times \frac{\pi}{6}$ $= \frac{\pi}{3\sqrt{3}}$ • 3 – correct solution • 2 – applies the substitution to express the integral in terms of t • 1 – applies the substitution to partially change the integral to an integral in terms of t

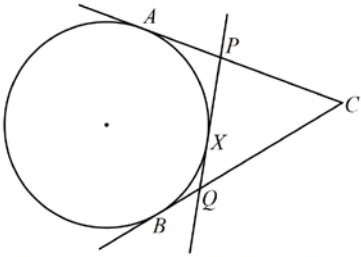
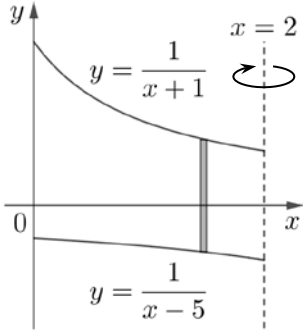
Question 12

Sample solution	Suggested marking criteria
(a) (i) $\sum \alpha = -1$ $\sum \alpha\beta = 0$ $\sum \alpha^2 = (\sum \alpha)^2 - 2\sum \alpha\beta$ $= (-1)^2 - 2 \times 0$ $= 1$ (ii) α, β and γ are roots of $x^3 + x^2 + 2 = 0$ $\alpha^3 + \alpha^2 + 2 = 0$ $\beta^3 + \beta^2 + 2 = 0$ $\gamma^3 + \gamma^2 + 2 = 0 +$ $(\alpha^3 + \beta^3 + \gamma^3) + (\alpha^2 + \beta^2 + \gamma^2) + 6 = 0$ $(\alpha^3 + \beta^3 + \gamma^3) + 1 + 6 = 0$ $\alpha^3 + \beta^3 + \gamma^3 = -7$ (iii) $\alpha^4 + \alpha^3 + 2\alpha = 0$ $\beta^4 + \beta^3 + 2\alpha = 0$ $\gamma^4 + \gamma^3 + 2\alpha = 0 +$ $(\alpha^4 + \beta^4 + \gamma^4) + (\alpha^3 + \beta^3 + \gamma^3) + 2(\alpha + \beta + \gamma) = 0$ $(\alpha^4 + \beta^4 + \gamma^4) + (-7) + 2 \times (-1) = 0$ $\alpha^4 + \beta^4 + \gamma^4 = 9$	• 2 – correct solution • 1 – finds $\sum \alpha$ • 2 – correct solution • 1 – forms an expression involving $\sum \alpha^3$ and $\sum \alpha^2$ • 2 – correct solution • 1 – forms an expression involving $\sum \alpha^3$, $\sum \alpha^2$ and $\sum \alpha$
(b) (i)  (ii) For w to lie in the region in part (i) and be of minimum modulus, w must be in the position as shown below:  By inspection, $w = \frac{1}{2} - \frac{1}{2}i$.	• 3 – correct region • 2 – correct boundary • 1 – correctly sketches one of the two regions of the given inequalities • 1 – correct answer

Question 12 (continued)

Sample solution		Suggested marking criteria
(c)	(i) $\frac{13!}{2!2!3!} = 259459200$ ways	• 1 – correct answer
	(ii) There are $\frac{7!}{2!2!3!}$ ways of arranging the seven repeated letter, then treating this as a group, there are $7!$ ways of arranging this group of repeated letter and the 6 remaining letters. Therefore, there are $\frac{7!}{2!2!3!} \times 7! = 1058400$ different arrangements.	• 2 – correct solution • 1 – correctly arranges the repeated letters
	(iii) Let v be a vowel and c be a consonant of the word BAULKHAMHILLS, start with a string of the nine consonants, like so: $_ c _ c _ c _ c _ c _ c _ c _ c _ c _$ With this string of consonants, there are ten spaces, choose any four of them to insert the vowels A, A, I and U in that order. Taking into account the number of possible arrangements of the consonants, this can be done in $\frac{9!}{2!3!} \times {}^{10}C_4 = 6350400$ ways.	• 2 – correct solution • 1 – considers some of the valid cases – progress towards solution using valid reasoning

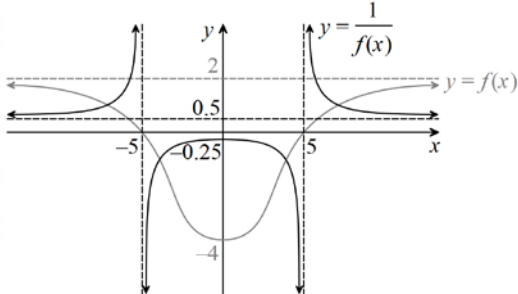
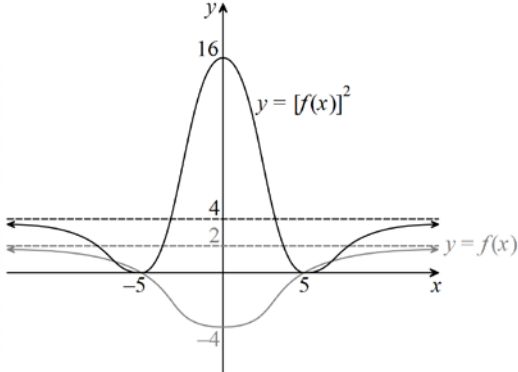
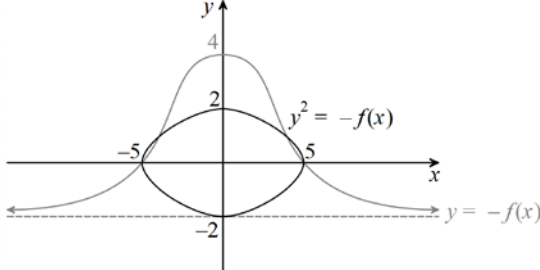
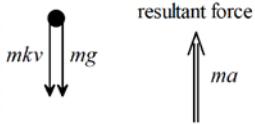
Question 13

Sample solution	Suggested marking criteria
(a)  Let X be the point of contact of the third tangent. $CA = CB$ (tangents from an external point are equal in length) $PA = PX$ (tangents from an external point are equal in length) $QB = QX$ (tangents from an external point are equal in length) The perimeter of $\triangle CPQ = CP + PQ + CQ$ $= CP + (PX + QX) + CQ$ $= (CP + PA) + (QB + CQ) \text{ (since } PX = PA \text{ and } QX = QB)$ $= CA + CB, \text{ which is independent of } PQ.$	• 2 – correct proof • 1 – one correct circle geometry reasoning towards the correct proof
(b) (i)  $\Delta V = 2\pi(2-x) \left(\frac{1}{x+1} - \frac{1}{x-5} \right) \Delta x$ $= 2\pi(2-x) \left[\frac{x-5-(x+1)}{(x+1)(x-5)} \right] \Delta x$ $= 2\pi(2-x) \left(\frac{-6}{x^2-4x-5} \right) \Delta x$ $= 12\pi \left(\frac{x-2}{x^2-4x-5} \right) \Delta x$	• 2 – correct solution • 1 – finds a correct expression for ΔV
(ii) $\Delta V = 12\pi \left(\frac{x-2}{x^2-4x-5} \right) \Delta x$ $V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^2 12\pi \left(\frac{x-2}{x^2-4x-5} \right) \Delta x$ $= \int_0^2 12\pi \left(\frac{x-2}{x^2-4x-5} \right) dx$ $= 6\pi \int_0^2 \left(\frac{2x-4}{x^2-4x-5} \right) dx$ $= 6\pi \left[\ln x^2-4x-5 \right]_0^2$ $= 6\pi (\ln -9 - \ln -5)$ $= 6\pi \ln \frac{9}{5} \text{ cubic units}$	• 3 – correct solution • 2 – correct primitive • 1 – establishes correct sum of infinitesimals
(c) (i) The tangent at T has equation $x + t^2 y = 2ct$. When $y = 0, x = 2ct, \therefore P = (2ct, 0)$ When $x = 0, y = \frac{2c}{t}, \therefore Q = \left(0, \frac{2c}{t} \right)$	• 2 – correct solution • 1 – correctly finds P or Q

Question 13 (continued)

Sample solution			Suggested marking criteria	
(c)	(ii)	$xy = c^2$ $y = \frac{c^2}{x}$ $\frac{dy}{dx} = \frac{-c^2}{x^2}$ At T , $\frac{dy}{dx} = -\frac{c^2}{(ct)^2}$ $= -\frac{1}{t^2}$	Equation of normal at T : $y - \frac{c}{t} = t^2(x - ct)$ $y - \frac{c}{t} = t^2x - ct^3$ $ty - c = t^3x - ct^4$ $ct^4 - c = t^3x - ty$ $t^3x - ty = c(t^4 - 1)$	• 2 – correct solution • 1 – correct gradient for the normal
	(iii)	$t^3x - ty = c(t^4 - 1)$ $t^3x - tx = c(t^2 + 1)(t^2 - 1)$ $xt(t^2 - 1) = c(t^2 + 1)(t^2 - 1)$ $xt = c(t^2 + 1)$ $x = \frac{c}{t}(t^2 + 1)$		• 2 – correct solution • 1 – attempts to use simultaneous equations to find the x-coordinate of R
	(iv)	$d_{PR} = \sqrt{\left[\frac{c}{t}(t^2 + 1) - 2ct\right]^2 + \left[\frac{c}{t}(t^2 + 1)\right]^2}$ $= \sqrt{\left(ct + \frac{c}{t} - 2ct\right)^2 + \left(ct + \frac{c}{t}\right)^2}$ $= \sqrt{\left(-ct + \frac{c}{t}\right)^2 + \left(ct + \frac{c}{t}\right)^2}$ $= \sqrt{c^2t^2 - 2c^2 + \frac{c^2}{t^2} + c^2t^2 + 2c^2 + \frac{c^2}{t^2}}$ $= \sqrt{2c^2t^2 + \frac{2c^2}{t^2}}$ $d_{QR} = \sqrt{\left[\frac{c}{t}(t^2 + 1)\right]^2 + \left[\frac{c}{t}(t^2 + 1) - \frac{2c}{t}\right]^2}$ $= \sqrt{\left(ct + \frac{c}{t}\right)^2 + \left(ct + \frac{c}{t} - \frac{2c}{t}\right)^2}$ $= \sqrt{\left(ct + \frac{c}{t}\right)^2 + \left(ct - \frac{c}{t}\right)^2}$ $= \sqrt{c^2t^2 + 2c^2 + \frac{c^2}{t^2} + c^2t^2 - 2c^2 + \frac{c^2}{t^2}}$ $= \sqrt{2c^2t^2 + \frac{2c^2}{t^2}}$ Since $d_{PR} = d_{QR}$, ΔPQR is isosceles.	• 2 – correct solution • 1 – finds d_{PR} or d_{QR} – establishes T as the midpoint of PQ – calculates the gradient of PQ or QR	

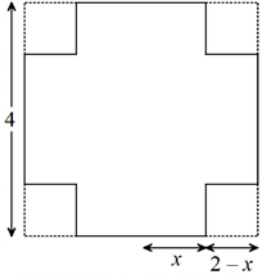
Question 14

Sample solution	Suggested marking criteria
(a) (i) 	• 2 – correct solution • 1 – significant progress towards solution
(ii) 	• 2 – correct solution • 1 – correctly identifies the turning points
(iii) 	• 2 – correct solution • 1 – sketches $y^2 = f(x)$ – sketches $y = \sqrt{-f(x)}$ – sketches $y = -\sqrt{-f(x)}$ – correctly positions the graph of $y^2 = f(x)$ but missing important features, or equivalent merit
(b) (i)  $ma = -mg - mkv$ $a = -(g + kv)$ $\frac{dv}{dt} = -(g + kv)$ $\frac{dt}{dv} = -\frac{1}{g + kv}$ $\int_0^T dt = \int_U^0 -\frac{1}{g + kv} dv$ $T = \frac{1}{k} \int_0^U \frac{k}{g + kv} dv$ $= \frac{1}{k} [\ln(g + kv)]_0^U$ $= \frac{1}{k} [\ln(g + kU) - \ln g]$ $= \frac{1}{k} \ln\left(\frac{g + kU}{g}\right)$	• 2 – correct solution • 1 – correctly integrates $-\frac{1}{g + kv}$

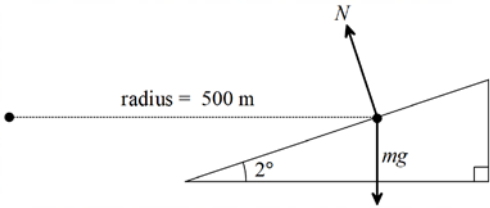
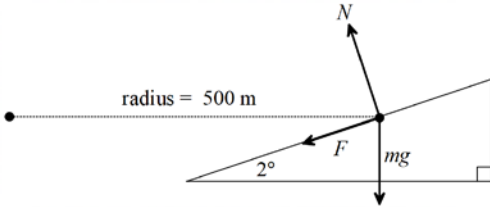
Question 14 (continued)

Sample solution		Suggested marking criteria
(b)	(ii) $a = -(g + kv)$ $v \frac{dv}{dx} = -(g + kv)$ $\frac{dv}{dx} = -\frac{g + kv}{v}$ $\frac{dx}{dv} = -\frac{v}{g + kv}$	• 1 – correct solution
	(iii) $\frac{dx}{dv} = -\frac{v}{g + kv}$ $\int_0^H dx = \int_U^0 -\frac{v}{g + kv} dv$ $H = \frac{1}{k} \int_0^U \frac{kv}{g + kv} dv$ $= \frac{1}{k} \int_0^U \frac{g + kv - g}{g + kv} dv$ $= \frac{1}{k} \int_0^U \left(1 - \frac{g}{g + kv} \right) dv$ $= \frac{1}{k} \int_0^U \left(1 - \frac{g}{k} \times \frac{k}{g + kv} \right) dv$ $= \frac{1}{k} \left[v - \frac{g}{k} \ln(g + kv) \right]_0^U$ $= \frac{1}{k} \left\{ \left[U - \frac{g}{k} \ln(g + kU) \right] - \left[0 - \frac{g}{k} \ln g \right] \right\}$ $= \frac{1}{k} \left[U - \frac{g}{k} \ln \left(\frac{g + kU}{g} \right) \right]$ $= \frac{1}{k} (U - gT)$ $kH = U - gT$ $U = kH + gT$	• 2 – correct solution • 1 – correctly integrates $-\frac{v}{g + kv}$

Question 14 (continued)

Sample solution	Suggested marking criteria
(c) (i)  $y = 4 - x^2$ $x^2 = 4 - y$ $x = \sqrt{4 - y}$ Area of slice = $4^2 - 4 \times (2 - x)^2$ $= 16 - 4 \times (2 - \sqrt{4 - y})^2$	• 1 – correct solution
(ii) $\Delta V = 4^2 - 4 \times (2 - \sqrt{4 - y})^2 \Delta y$ $V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^4 \left[16 - 4 \times (2 - \sqrt{4 - y})^2 \right] \Delta y$ $= \int_0^4 \left[16 - 4 \times (2 - \sqrt{4 - y})^2 \right] dy$ $= \int_0^4 \left[16 - 4 \times (4 - 4\sqrt{4 - y} + 4 - y) \right] dy$ $= \int_0^4 \left[16 - 4 \times (8 - 4\sqrt{4 - y} - y) \right] dy$ $= \int_0^4 (16 - 32 + 16\sqrt{4 - y} + 4y) dy$ $= \int_0^4 (-16 + 16\sqrt{4 - y} + 4y) dy$ $= \int_0^4 \left(-16 + 16(4 - y)^{\frac{1}{2}} + 4y \right) dy$ $= \left[-16y + \frac{16(4 - y)^{\frac{3}{2}}}{-\left(\frac{3}{2}\right)} + 2y^2 \right]_0^4$ $= \left[-16y - \frac{32(4 - y)^{\frac{3}{2}}}{3} + 2y^2 \right]_0^4$ $= (-64 + 32) + \left(\frac{32 \times 4^{\frac{3}{2}}}{3} \right)$ $= 53\frac{1}{3} \text{ cubic units}$	• 3 – correct solution • 2 – correct primitive • 1 – correct integral

Question 15

Sample solution	Suggested marking criteria
(a) (i)  radius = 500 m 2° N mg Resolving forces vertically: $N \cos 2^\circ - mg = 0$ $N \cos 2^\circ = 50000$ Resolving forces horizontally: $N \sin 2^\circ = \frac{mv^2}{r}$ $N \sin 2^\circ = \frac{5000v^2}{500}$ $= 10v^2$ $\frac{N \sin 2^\circ}{N \cos 2^\circ} = \frac{10v^2}{50000}$ $\tan 2^\circ = \frac{v^2}{5000}$ $v = \sqrt{5000 \tan 2^\circ}$ $= 13.21 \text{ m/s (2 d.p.)}$	• 2 – correct solution • 1 – resolves the forces vertically and horizontally, or equivalent merit
(ii)  radius = 500 m 2° N mg F Resolving forces vertically: $N \cos 2^\circ - F \sin 2^\circ - mg = 0$ $N \cos 2^\circ - F \sin 2^\circ = 50000$ Resolving forces horizontally: $N \sin 2^\circ + F \cos 2^\circ = \frac{mv^2}{r}$ $N \sin 2^\circ + F \cos 2^\circ = \frac{5000 \times 20^2}{500}$ Eliminating $N \sin 2^\circ \cos 2^\circ$: $N \sin 2^\circ \cos 2^\circ - F \sin^2 2^\circ = 50000 \sin 2^\circ$ $N \sin 2^\circ \cos 2^\circ + F \cos^2 2^\circ = 4000 \cos 2^\circ$ $F \cos^2 2^\circ + F \sin^2 2^\circ = 4000 \cos 2^\circ - 50000 \sin 2^\circ$ $F = 2252.6 \text{ N (1 d.p.)}$	• 3 – correct solution • 2 – resolves forces vertically and horizontally, or equivalent merit • 1 – resolves forces in one direction
(b) (i) $\tan\left(A + \frac{\pi}{2}\right) = \cot\left[\frac{\pi}{2} - \left(A + \frac{\pi}{2}\right)\right]$ $= \cot(-A)$ $= \frac{1}{\tan(-A)}$ $= \frac{1}{-\tan A} \text{ (tan x is an odd function)}$ $= -\cot A$	• 1 – correct solution

Question 15 (continued)

Sample solution	Suggested marking criteria
(b) (ii) Let $S(n)$ be the statement that $\tan\left[(2n+1)\frac{\pi}{4}\right] = (-1)^n$. Show $S(1)$ is true: $\begin{aligned} \text{LHS} &= \tan\left(\frac{3\pi}{4}\right) & \text{RHS} &= (-1)^1 \\ &= -\tan\frac{\pi}{4} & &= -1 \\ &= -1 \end{aligned}$ Since $\text{LHS} = \text{RHS}$, $\therefore S(1)$ is true. Assume $S(k)$ is true: i.e. $\tan\left[(2k+1)\frac{\pi}{4}\right] = (-1)^k$ Prove that $S(k+1)$ is true: i.e. $\tan\left[(2k+3)\frac{\pi}{4}\right] = (-1)^{k+1}$ $\begin{aligned} \text{LHS} &= \tan\left[(2k+3)\frac{\pi}{4}\right] \\ &= \tan\left[(2k+1)\frac{\pi}{4} + 2 \times \frac{\pi}{4}\right] \\ &= \tan\left[(2k+1)\frac{\pi}{4} + \frac{\pi}{2}\right] \\ &= -\cot\left[(2k+1)\frac{\pi}{4}\right] \\ &= -\frac{1}{\tan\left[(2k+1)\frac{\pi}{4}\right]} \\ &= -\frac{1}{(-1)^k} \\ &= \frac{-1}{(-1)^k} \times \frac{(-1)^k}{(-1)^k} \\ &= \frac{(-1)^{k+1}}{(-1)^{2k}} \\ &= \frac{(-1)^{k+1}}{[(-1)^2]^k} \\ &= \frac{(-1)^{k+1}}{1^k} \\ &= (-1)^{k+1} \\ &= \text{RHS} \end{aligned}$ $\therefore S(k+1)$ is true if $S(k)$ is assumed true. Since $S(1)$ is proven true, then by the principle of mathematical induction, $S(n)$ is true for all integers $n \geq 1$.	• 3 – correct solution • 2 – uses the inductive hypothesis to attempt to prove the result inductively • 1 – showing the result to be true for $n = 1$

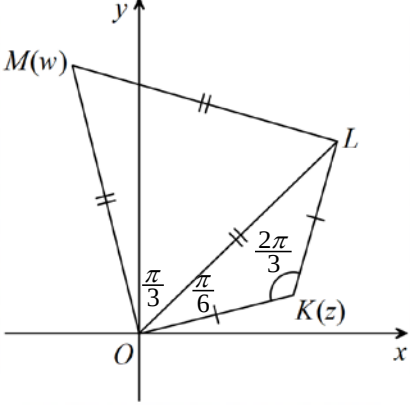
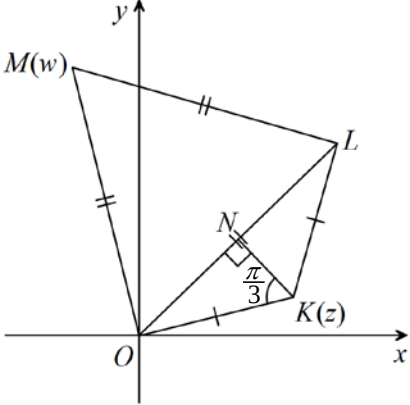
Question 15 (continued)

Sample solution	Suggested marking criteria
(c) (i) $ \begin{aligned} I_0 &= \int_0^1 \frac{1}{\sqrt{x+1}} dx \\ &= \int_0^1 (x+1)^{-\frac{1}{2}} dx \\ &= \left[2(x+1)^{\frac{1}{2}} \right]_0^1 \\ &= 2 \times 2^{\frac{1}{2}} - 2 \\ &= 2\sqrt{2} - 2 \end{aligned} $	• 1 – correct solution
(ii) $ \begin{aligned} I_{n-1} + I_n &= \int_0^1 \frac{x^{n-1}}{\sqrt{x+1}} dx + \int_0^1 \frac{x^n}{\sqrt{x+1}} dx \\ &= \int_0^1 \frac{x^{n-1}}{\sqrt{x+1}} + \frac{x^n}{\sqrt{x+1}} dx \\ &= \int_0^1 \frac{x^{n-1} + x^n}{\sqrt{x+1}} dx \\ &= \int_0^1 \frac{x^{n-1}(1+x)}{\sqrt{x+1}} dx \\ &= \int_0^1 x^{n-1} \sqrt{x+1} dx \end{aligned} $	• 1 – correct solution
(iii) Using integration by parts, $ \begin{aligned} I_{n-1} + I_n &= \left[\frac{x^n \sqrt{x+1}}{n} \right]_0^1 - \int_0^1 \frac{x^n}{2n\sqrt{x+1}} dx & u &= \sqrt{x+1} & du &= \frac{dx}{2\sqrt{x+1}} \\ &= \frac{\sqrt{2}}{n} - \frac{1}{2n} \int_0^1 \frac{x^n}{\sqrt{x+1}} dx & v &= \frac{x^n}{n} & dv &= x^{n-1} dx \\ &= \frac{\sqrt{2}}{n} - \frac{1}{2n} I_n \\ I_n + \frac{1}{2n} I_n &= \frac{\sqrt{2}}{n} - I_{n-1} \\ I_n \left(1 + \frac{1}{2n} \right) &= \frac{\sqrt{2}}{n} - I_{n-1} \\ I_n (2n+1) &= 2\sqrt{2} - 2nI_{n-1} \\ I_n &= \frac{2\sqrt{2} - 2nI_{n-1}}{2n+1} \end{aligned} $	• 2 – correct solution • 1 – attempts to use integration by parts towards the required result

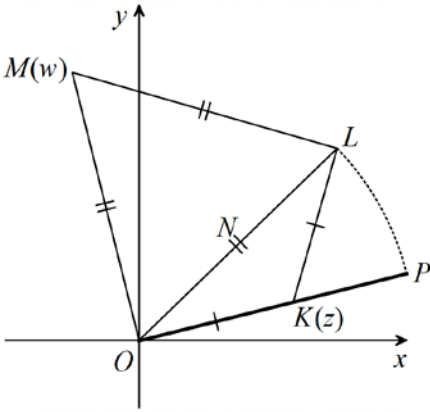
Question 15 (continued)

Sample solution	Suggested marking criteria
(c) (iv) $\int_0^1 \frac{x^2}{\sqrt{x+1}} dx = I_2$ $= \frac{2\sqrt{2} - 4I_1}{5}$ $= \frac{2\sqrt{2} - 4 \times \left(\frac{2\sqrt{2} - 2I_0}{3} \right)}{5}$ $= \frac{2\sqrt{2} - 4 \times \left[\frac{2\sqrt{2} - 2 \times (2\sqrt{2} - 2)}{3} \right]}{5}$ $= \frac{2\sqrt{2} - 4 \times \left[\frac{2\sqrt{2} - 4\sqrt{2} + 4}{3} \right]}{5}$ $= \frac{2\sqrt{2} - 4 \times \left[\frac{4 - 2\sqrt{2}}{3} \right]}{5}$ $= \frac{6\sqrt{2} - 4 \times (4 - 2\sqrt{2})}{15}$ $= \frac{6\sqrt{2} - 16 + 8\sqrt{2}}{15}$ $= \frac{14\sqrt{2} - 16}{15}$	• 2 – correct solution • 1 – one correct application of the reduction formula – finds I_1

Question 16

Sample solution	Suggested marking criteria
(a) (i)  In $\triangle OKL$, $\angle KOL = \angle KLO$ (angles opposite equal sides of a triangle are equal) $\angle KOL + \angle KLO + \angle OKL = \pi$ (angle sum of $\triangle OKL$ is π radians) $\angle KOL + \angle KOL + \frac{2\pi}{3} = \pi$ $2\angle KOL = \frac{\pi}{3}$ $\angle KOL = \frac{\pi}{6}$ In $\triangle OML$, $\angle MOL = \frac{\pi}{3}$ (angles in an equilateral triangle) $\angle MOK = \angle MOL + \angle KOL$ $= \frac{\pi}{3} + \frac{\pi}{6}$ $= \frac{\pi}{2}$	• 1 – correct solution
(ii)  Since $\triangle OKL$ is isosceles, by symmetry, if KN is an altitude of the triangle, N bisect OL and $\angle OKN = \frac{\pi}{3}$. $\sin \frac{\pi}{3} = \frac{ \overline{ON} }{ \overline{OK} }$ $\frac{\sqrt{3}}{2} = \frac{\frac{1}{2} \overline{OL} }{ z }$ $\sqrt{3} = \frac{ \overline{OL} }{ z }$ $ \overline{OL} = \sqrt{3} \times z $	• 2 – correct solution • 1 – uses trigonometry to attempt to show the required result

Question 16 (continued)

Sample solution	Suggested marking criteria
(a) (iii)  Suppose $\overrightarrow{OP}$ is the vector in the direction of $\overrightarrow{OK}$ with length equal to $\overrightarrow{OL}$. Since $\overrightarrow{OL} = \sqrt{3} \times z$, therefore $\overrightarrow{OP} = \sqrt{3} \times z$, i.e. $\overrightarrow{OP} = \sqrt{3} \times \overrightarrow{OK}$. $\overrightarrow{OM} = i\overrightarrow{OP}$ $\overrightarrow{OM} = i\sqrt{3} \times \overrightarrow{OK}$ $w = i\sqrt{3} \times z$ $w^2 = -3z^2$ $3z^2 + w^2 = 0$	• 2 – correct solution • 1 – expresses w in terms of z
(b) (i) $y = \frac{1}{2}(e^x + e^{-x})$ $\frac{dy}{dx} = \frac{1}{2}(e^x - e^{-x})$ $1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left[\frac{1}{4}(e^x - e^{-x})^2\right]$ $= 1 + \frac{(e^x)^2 - 2 + (e^{-x})^2}{4}$ $= \frac{4 + (e^x)^2 - 2 + (e^{-x})^2}{4}$ $= \frac{(e^x)^2 + 2 + (e^{-x})^2}{4}$ $= \frac{(e^x + e^{-x})^2}{4}$	• 2 – correct solution • 1 – finds $\frac{dy}{dx}$
(ii) $L = \int_0^1 \sqrt{\frac{(e^x + e^{-x})^2}{4}} dx$ $= \int_0^1 \frac{e^x + e^{-x}}{2} dx$ $= \frac{1}{2} [e^x - e^{-x}]_0^1$ $= \frac{1}{2} [(e - e^{-1}) - (e^0 - e^0)]$ $= \frac{1}{2} \left(e - \frac{1}{e}\right) \text{ units}$	• 2 – correct solution • 1 – correct integration

Question 16 (continued)

Sample solution		Suggested marking criteria
(c)	(i) $P(x) = x^n + px - q$ $P'(x) = nx^{n-1} + p$ $P'(\alpha) = 0$ $n\alpha^{n-1} + p = 0$ $n\alpha^{n-1} = -p$ $\alpha^{n-1} = -\frac{p}{n}$	• 2 – correct solution • 1 – finds $P'(\alpha)$
	(ii) $P'(\alpha) = 0$ $n\alpha^{n-1} + p = 0$ $n\alpha^n + p\alpha = 0$ $P(x) = x^n + px - q$ $P(\alpha) = 0$ $\alpha^n + p\alpha - q = 0$ $\alpha^n + p\alpha = q$ Solving simultaneously: $(\alpha^n + p\alpha) - (n\alpha^n + p\alpha) = q$ $\alpha^n - n\alpha^n = q$ $\alpha^n(1-n) = q$ $\alpha^n = \frac{q}{1-n}$	$(\alpha^{n-1})^n = \left(-\frac{p}{n}\right)^n$ $(\alpha^n)^{n-1} = \left(\frac{q}{1-n}\right)^{n-1}$ $\left(\frac{q}{1-n}\right)^{n-1} = \left(-\frac{p}{n}\right)^n$ $\left(\frac{-q}{n-1}\right)^{n-1} = (-1)^n \left(\frac{p}{n}\right)^n$ $(-1)^{n-1} \left(\frac{q}{n-1}\right)^{n-1} = (-1)^n \left(\frac{p}{n}\right)^n$ $\left(\frac{q}{n-1}\right)^{n-1} = -\left(\frac{p}{n}\right)^n$ $\left(\frac{p}{n}\right)^n + \left(\frac{q}{n-1}\right)^{n-1} = 0$ • 3 – correct solution • 2 – substitutes α, α^n or α^{n-1} into an appropriate expression / equation and attempts to simply • 1 – uses $P(\alpha) = 0$ – finds α or α^n
	(iii) $\alpha = \frac{\alpha^n}{\alpha^{n-1}}$ $= \frac{\left(\frac{q}{1-n}\right)}{\left(-\frac{p}{n}\right)}$ $= \frac{-qn}{p(1-n)}$ $= \frac{qn}{p(n-1)}$	• 1 – correct solution